



Exp :4 Write the Python Program for Cript-Arithmetic Problem

Aim:

To Write the Python Program for **Cript-Arithmetic Problem**

Algorithm:

Step 1:Write down the arithmetic problem in full, including any carry-overs.

Step 2: Replace each letter with a unique digit from 0 to 9.

Step 3:Generate all possible combinations of digits for each letter, subject to any constraints ' from the carry-overs and the rules of arithmetic.

Step 4: For each combination of digits, evaluate the arithmetic problem and check if it satisfies the given constraints.

Step 5:If a solution is found, output the values of the letters and the arithmetic problem, and exit the algorithm.

Step 6:If no solution is found, output a message indicating that no solution exists.

Program:

```
import itertools
```

```
import re
```

```
def solve_cryptarithmic(puzzle):
```

```
    words = re.findall(r'[A-Z]+' , puzzle)
```

```
    if not words or len(words) < 3:
```

```
        return None
```

```
    letters = set(''.join(words))
```

```
    if len(letters) > 10:
```

```
        return None
```

```
    letters = list(letters)
```

```
    first_letters = {word[0] for word in words}
```